U.S. PhDs before and after ChatGPT

Research doctorate recipients at U.S. institutions | 2015-2025 | NCSES Survey of Earned Doctorates

Question and scope

How did annual research doctorate counts change around the November 30, 2022 public launch of ChatGPT? This is a descriptive timing comparison, with a mathematics and statistics focus. It does not estimate an effect caused by ChatGPT.

Survey year	All research doctorates	Mathematics and statistics	Math/stat share
2015	54,886	1,815	3.31%
2016	54,809	1,872	3.42%
2017	54,550	1,844	3.38%
2018	55,077	2,022	3.67%
2019	55,597	2,010	3.62%
2020	55,200	2,024	3.67%
2021	52,158	2,010	3.85%
2022	57,435	2,247	3.91%
2023	57,801	2,165	3.75%
2024	58,013	2,092	3.61%
2025	59,015	2,175	3.69%

Period averages

Mean annual all-field count: **54,964** in 2015-2022 versus **58,276** in 2023-2025 (+6.0%). Mathematics and statistics: **1,980** versus **2,144** (+8.3%). The windows have different lengths; these are simple averages, not adjusted forecasts.

How to read this

Survey year 2023 spans July 2022-June 2023 and straddles launch. A 2023-2025 graduation reflects admissions and research decisions made years earlier. The 2021 field taxonomy change complicates subfield trends; COVID, funding, and immigration also matter. Counts cannot show whether AI changed demand for a PhD.

Source: NCSES, Doctorate Recipients from U.S. Universities: 2025 Data Tables, NSF 26-326, Table 1-5 (Sept. 25, 2026). ncses.nsf.gov/surveys/earned-doctorates/2025. The accompanying repository contains the data, code, chart, and a fuller methodology.